

" 2023 and older "

Part I:

$$K = (0, i, j), K' = (0', i', j')$$

$$[0']_K = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, [i']_K = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, [j']_K = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$K \rightarrow$  right-oriented orthonormal sys.

$$\Delta ABC: [A]_K = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, [B]_K = \begin{bmatrix} -3 \\ 1 \end{bmatrix}, [C]_K = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

P1)  $\vec{BA}, \vec{CA}$  orthogonal to each other?  $\Leftrightarrow \langle \vec{BA}, \vec{CA} \rangle = 0$

$$\vec{BA} = A - B = \begin{bmatrix} 1 \\ 3 \end{bmatrix} - \begin{bmatrix} -3 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

$$\Leftrightarrow (4 \cdot 2) + (2 \cdot 2) = 0$$

$$\Leftrightarrow 12 = 0 \text{ false.}$$

$$\vec{CA} = A - C = \begin{bmatrix} 1 \\ 3 \end{bmatrix} - \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

$\Rightarrow$  not orthogonal.

P2) w.r. to  $K$ , Cartesian eq. for a line passing through C  
 $\nless 45^\circ$  with line AB.

$$\text{line: } ax + by + c = 0$$

$$C \in \text{line} \Rightarrow a \cdot (-1) + b \cdot 1 + c = 0$$

$$\Rightarrow -a + b + c = 0.$$

$$\vec{AB} = B - A = \begin{bmatrix} -4 \\ -2 \end{bmatrix} \Rightarrow \text{direction vector: } \underset{AB}{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

normal vector. direction vector = 0

$$m \cdot v = 0$$

$$\underset{AB}{m} \cdot (2, 1) = 0$$

$$(1, -2) \cdot (2, 1) = 2 - 2 = 0$$

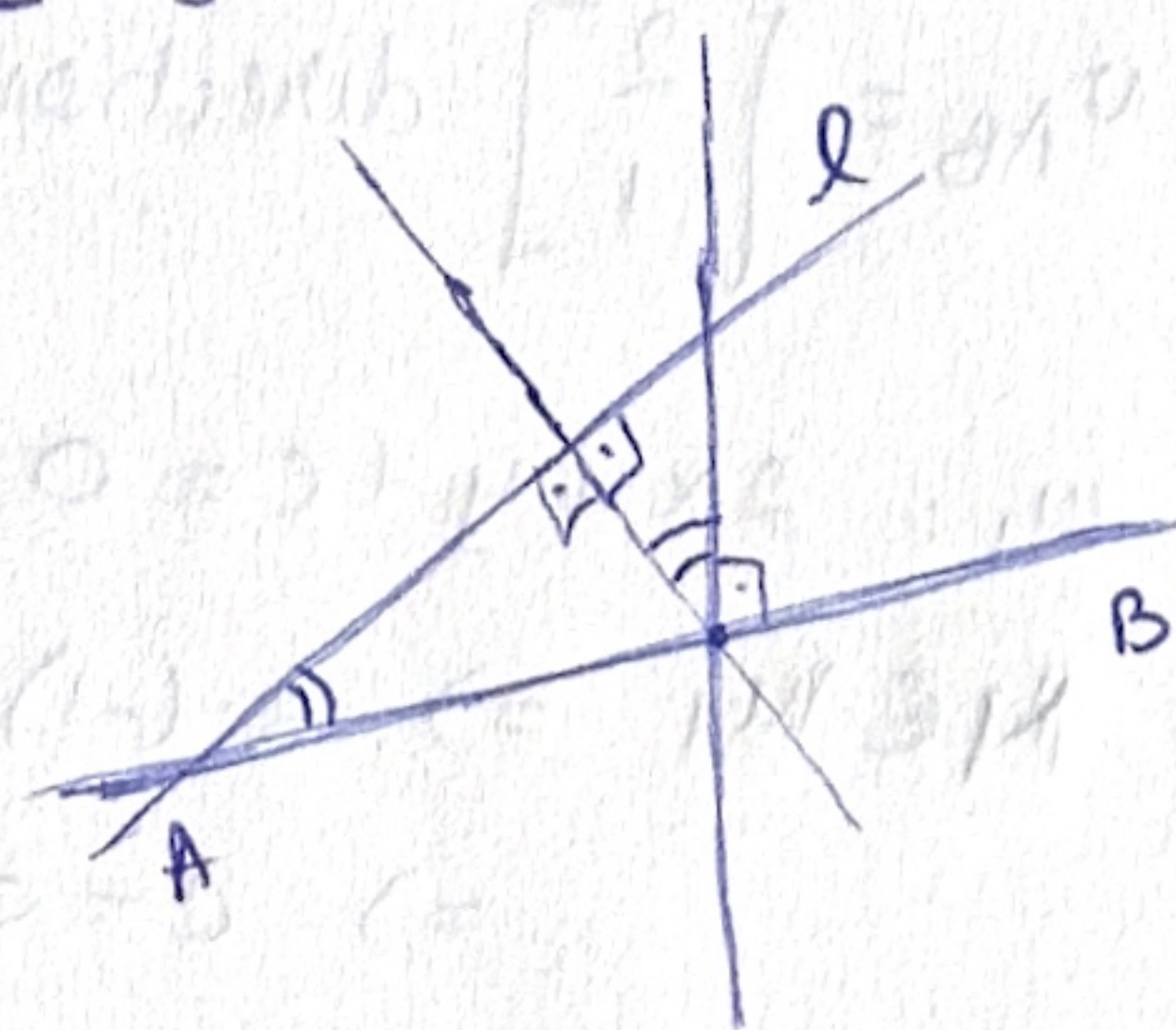
$$m_{AB} = (1, -2) \text{ normal vector.}$$

$$\|v\| = \sqrt{2^2 + 1^2} = \sqrt{5}$$

$$\|m_{AB}\| = \sqrt{1^2 + (-2)^2} = \sqrt{5}$$

$$m_1 = m_{AB} + v = \begin{bmatrix} 1 \\ -2 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

$$m_2 = m_{AB} - v = \begin{bmatrix} 1 \\ -2 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -3 \end{bmatrix} \sim \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$





$$a(x-x_c) + b(y-y_c) = 0$$

its normal vector  $(a, b)$ .

•  $m_1 = (3, -1)$  normal vector.

$$3(x+1) - 1 \cdot (y-1) = 0$$

$$3x + 3 - y + 1 = 0$$

$$\underline{3x - y + 4 = 0.}$$

•  $m_2 = (1, 3)$

$$1 \cdot (x+1) + 3 \cdot (y-1) = 0$$

$$x + 1 + 3y - 3 = 0$$

$$\underline{x + 3y - 2 = 0.}$$

P3) O - circumcenter of  $\triangle ABC$ .

$$\vec{AC} = C - A = \begin{bmatrix} -2 \\ -2 \end{bmatrix} \Rightarrow \vec{AC} \text{ has direction } (1, 1).$$

$$H = \frac{A+B}{2} = \left( \frac{1+(-3)}{2}, \frac{3+1}{2} \right) = (-1, 2)$$

$$\vec{AB} = B - A = \begin{bmatrix} -3 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} -4 \\ -2 \end{bmatrix}$$

$$v_{AB} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \text{ direction vector.}$$

$$m_1: 2x + y + c = 0$$

$$H \in m_1 \Rightarrow 2 \cdot (-1) + 2 + c = 0$$

$$\Rightarrow c = 0$$

$$\Rightarrow \boxed{m_1: 2x + y = 0}$$

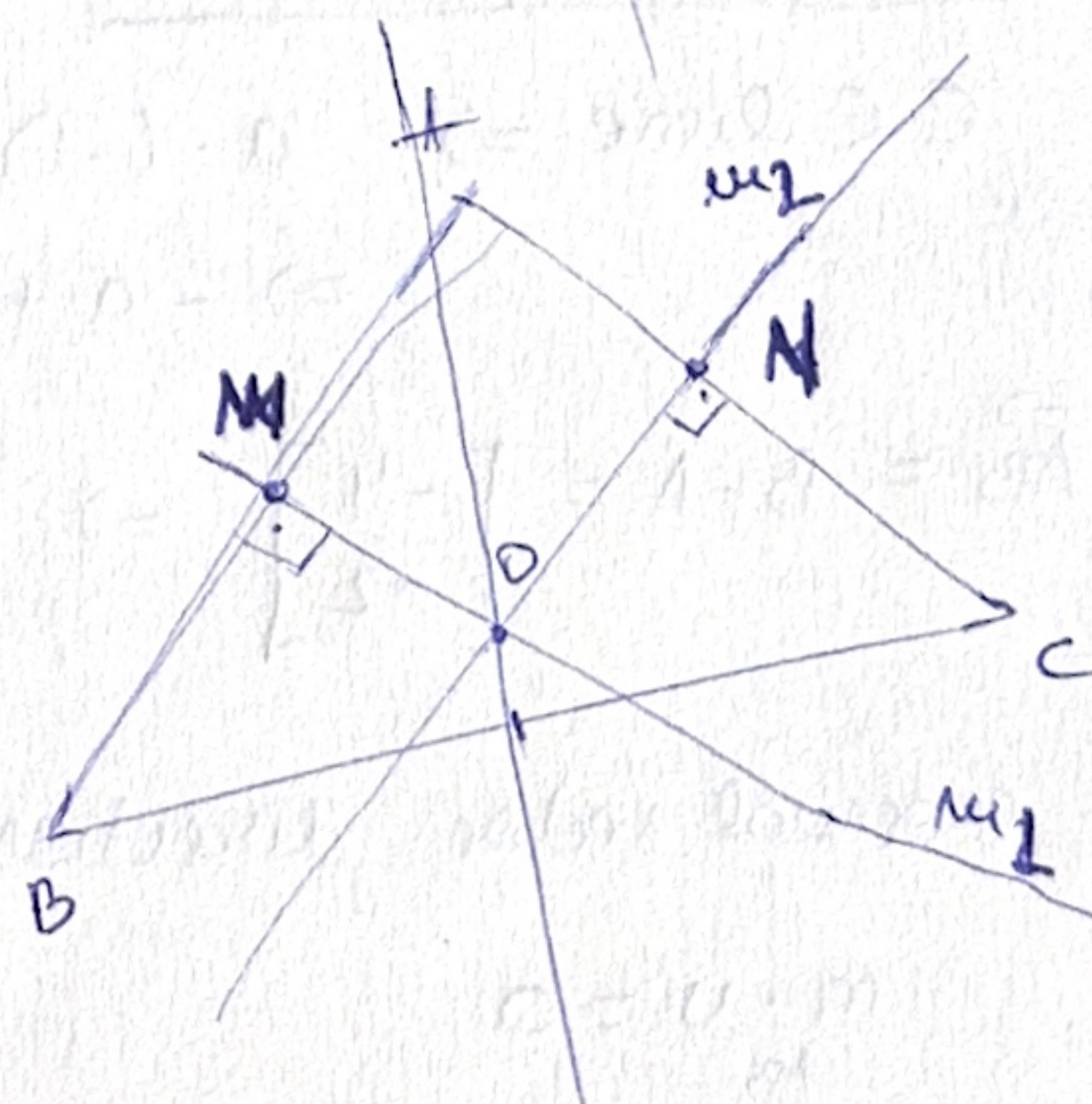
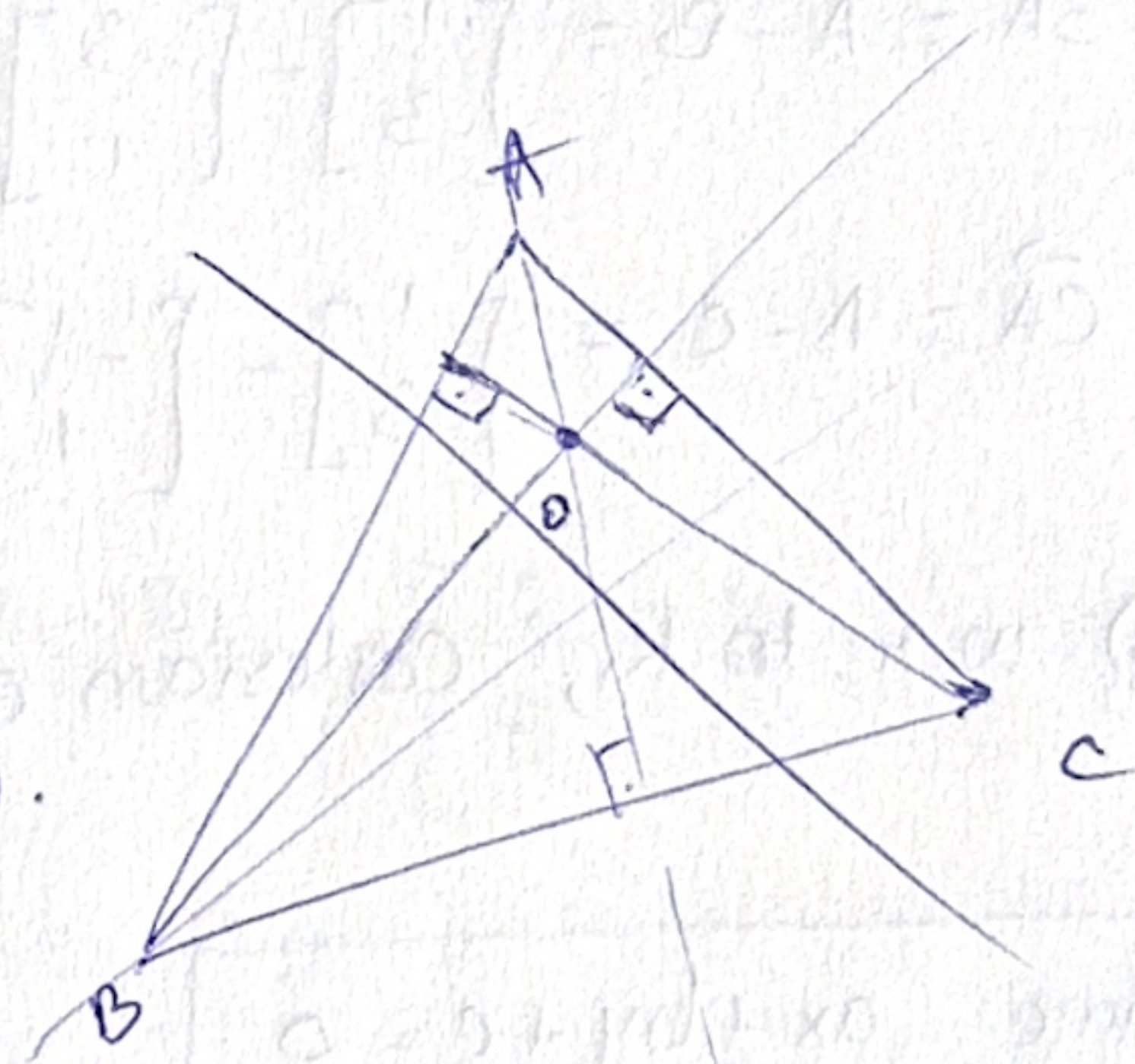
$$N = \frac{A+C}{2} = \left( \frac{1-1}{2}, \frac{3+1}{2} \right) = (0, 2)$$

$$m_2: 1 \cdot x + 1 \cdot y + c = 0$$

$$N \in m_2 \Rightarrow 1 \cdot 0 + 2 \cdot 2 + c = 0 \Rightarrow c = -2$$

$$\Rightarrow \underline{2y = -c}$$

$$\Rightarrow \boxed{m_2: x + y - 2 = 0}$$



$$\begin{cases} 2x + y = 0 \\ x + y - 2 = 0 \Rightarrow x = 2 - y \end{cases}$$

$$2(2 - y) + y = 0$$

$$4 - 2y + y = 0$$

$$y = 4$$

$$x = -2$$

$$\Rightarrow O(-2, 4).$$

(2)

(3)



Ph) AC w.r. to  $K'$ .

$$[A]_{K'} = M_{B'B} ([A]_K - [O']_K)$$

$$M_{B'B} = \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} \quad \begin{matrix} \downarrow & \downarrow \\ i' & j' \end{matrix} \quad \leadsto \quad M_{B'B} = \begin{bmatrix} \frac{2}{5} & \frac{1}{5} \\ -\frac{1}{5} & \frac{2}{5} \end{bmatrix}$$

$$\left[ \begin{array}{cc|cc} 2 & -1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{array} \right] \xrightarrow{2R_1 + R_2 \rightarrow R_2} \left[ \begin{array}{cc|cc} 2 & -1 & 1 & 0 \\ 5 & 0 & 2 & 1 \end{array} \right]$$

$$\xrightarrow{-2R_2 + R_1 \rightarrow R_1} \left[ \begin{array}{cc|cc} 0 & -5 & -3 & -1 \\ 1 & 2 & 0 & 1 \end{array} \right]$$

-4-1

$$\left[ \begin{array}{cc|cc} 2 & -1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{array} \right] \xrightarrow{2R_1 + R_2 \rightarrow R_2} \left[ \begin{array}{cc|cc} 2 & -1 & 1 & 0 \\ 5 & 0 & 2 & 1 \end{array} \right]$$

$$\left[ \begin{array}{cc|cc} 2 & -1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{array} \right] \xrightarrow{R_2 + R_1 \rightarrow R_2} \left[ \begin{array}{cc|cc} 2 & -1 & 1 & 0 \\ 3 & 1 & 1 & 1 \end{array} \right]$$

$$\left[ \begin{array}{cc|cc} 2 & -1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[ \begin{array}{cc|cc} 1 & 2 & 0 & 1 \\ 2 & -1 & 1 & 0 \end{array} \right] \xrightarrow{R_2 \leftarrow R_2 - 2R_1}$$

$$\left[ \begin{array}{cc|cc} 1 & 2 & 0 & 1 \\ 0 & -5 & 1 & -2 \end{array} \right] \xrightarrow{R_2 \leftarrow -\frac{1}{5}R_2} \left[ \begin{array}{cc|cc} 1 & 2 & 0 & 1 \\ 0 & 1 & -\frac{1}{5} & \frac{2}{5} \end{array} \right]$$

-1-2.2

$$\xrightarrow{R_1 \leftarrow R_1 - R_2} \left[ \begin{array}{cc|cc} 1 & 0 & \frac{2}{5} & \frac{1}{5} \\ 0 & 1 & -\frac{1}{5} & \frac{2}{5} \end{array} \right]$$

$M_{B'B}^{-1}$

$$[A]_{K'} = \begin{bmatrix} \frac{2}{5} & \frac{1}{5} \\ -\frac{1}{5} & \frac{2}{5} \end{bmatrix} \cdot \left( \begin{bmatrix} 1 \\ 3 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$$

$$= \begin{bmatrix} \frac{2}{5} & \frac{1}{5} \\ -\frac{1}{5} & \frac{2}{5} \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{2}{5} + \frac{2}{5} \\ -\frac{1}{5} + \frac{4}{5} \end{bmatrix} = \begin{bmatrix} \frac{4}{5} \\ \frac{3}{5} \end{bmatrix}$$



$$[c]_{k'} = M_{B'B} \cdot ([c]_k - [0]_k)$$

$$= \begin{bmatrix} \frac{2}{5} & \frac{1}{5} \\ -\frac{1}{5} & \frac{2}{5} \end{bmatrix} \cdot \left( \begin{bmatrix} -1 \\ 1 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$$

$$= \begin{bmatrix} \frac{2}{5} & \frac{1}{5} \\ -\frac{1}{5} & \frac{2}{5} \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{2}{5} \\ \frac{1}{5} \end{bmatrix}$$

$$\vec{CA} = A - C = \begin{bmatrix} \frac{4}{5} \\ \frac{3}{5} \end{bmatrix} - \begin{bmatrix} -\frac{2}{5} \\ \frac{1}{5} \end{bmatrix} = \begin{bmatrix} \frac{6}{5} \\ \frac{2}{5} \end{bmatrix} \sim \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

$\Rightarrow$  director vector :  $(3, 1)$ .

$\Rightarrow$  normal vector :  $(1, -3)$ .

$$(1, -3) \cdot (3, 1) = 3 - 3 = 0.$$

$$Ac \equiv ax + by + c = 0.$$

$$A' \left( \frac{4}{5}, \frac{3}{5} \right), \quad C' \left( -\frac{2}{5}, \frac{1}{5} \right)$$

$$\vec{CA} = (3, 1) \text{ dir. vector}$$

$$(x', y') = \left( \frac{4}{5}, \frac{3}{5} \right) + t \cdot (3, 1)$$

$\downarrow$   
A

$\downarrow$   
dir.  
vector

$$\Rightarrow \begin{cases} x' = \frac{4}{5} + 3t \\ y' = \frac{3}{5} + t \end{cases}$$

$$a = 1$$

$$b = -3.$$

$$x' - 3y' + c = 0$$

$$\Rightarrow \frac{4}{5} - 3 \cdot \frac{3}{5} + c = 0$$

$$\Rightarrow -\frac{5}{5} + c = 0$$

$$\Rightarrow c = 1$$

$$\Rightarrow \underline{x' - 3y' + 1 = 0}$$

$$Ac: x' - 3y' + 1 = 0.$$



Part II.  $K = (O, i, j, k)$  right-oriented orthon. sys.

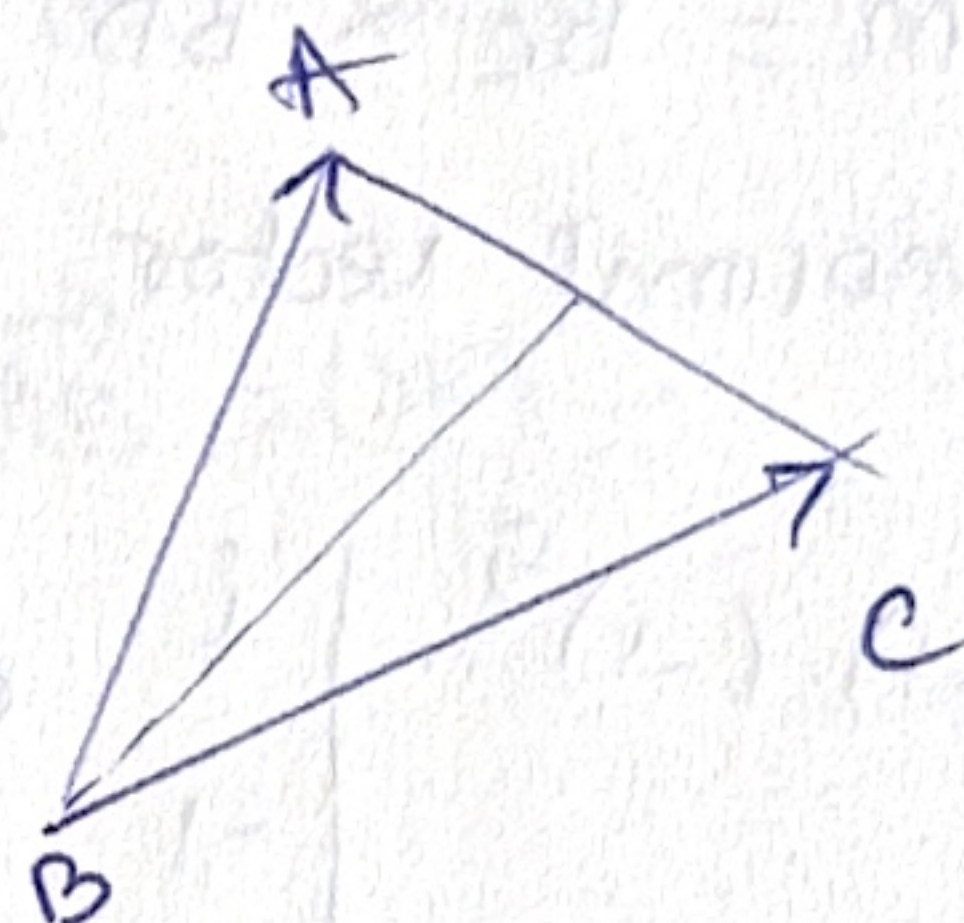
ABCD tetrahedron.

$$[A]_K = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, [B]_K = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, [C]_K = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}, [D]_K = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

$$K' = (B, \vec{BA}, \vec{AC}, \vec{CD}).$$

75)  $\vec{BA} = A - B = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$

$$\vec{BC} = C - B = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$



$$\vec{d} = \frac{\vec{BA}}{\|\vec{BA}\|} + \frac{\vec{BC}}{\|\vec{BC}\|} = 1 \cdot \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} + \frac{1}{\sqrt{3}} \cdot \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ -1 \end{bmatrix}$$

$$\|\vec{BA}\| = \sqrt{0^2 + 0^2 + (-1)^2} = 1.$$

$$\|\vec{BC}\| = \sqrt{(-1)^2 + 1^2 + 1^2} = \sqrt{3}.$$

line through B with direction  $\vec{d}$ :

$$\begin{cases} x = 1 - t \cdot \frac{1}{\sqrt{3}} \\ y = 1 + t \cdot \frac{1}{\sqrt{3}} \\ z = 0 + t \cdot \left( \frac{1}{\sqrt{3}} - 1 \right) \end{cases}$$

$$\Rightarrow \begin{cases} x = 1 - t \\ y = 1 + t \\ z = t(1 - \sqrt{3}) \end{cases}$$

$$\Rightarrow (x, y, z) = (1, 1, 0) + t \cdot (-1, 1, 1 - \sqrt{3}).$$

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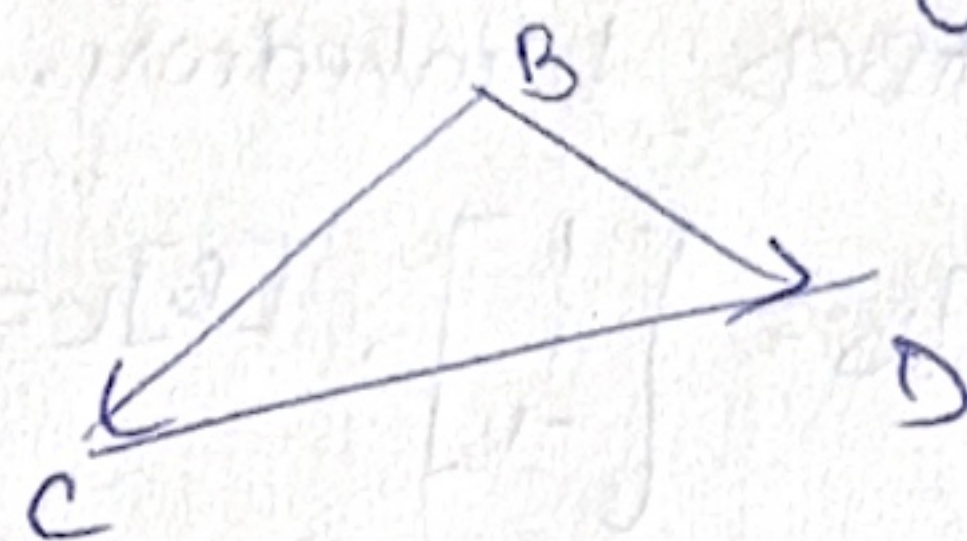
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P6) w.r. to  $K$ , orthogonal refl. on  $A$  in plane containing  $\triangle BCD$ .

$$\vec{BC} = C - B = (-1, 1, 1)$$

$$\vec{BD} = D - B = (0, -1, 0)$$



$$m = \vec{BC} \times \vec{BD} = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \times \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 1 & 1 \\ 0 & -1 & 0 \end{vmatrix} =$$

(normal vector)

$$= (-1) \cdot (-1) \hat{i} - (-1) \hat{j} + (-1) \hat{k} = \hat{i} + \hat{j} - \hat{k} = (1, 1, -1).$$

~~$a \cdot x + b$~~

$$B(1, 1, 0) \in (BCD), \quad m = (1, 0, 1).$$

$$1 \cdot (x - x_B) + 0 \cdot (y - y_B) + 1 \cdot (z - z_B) = 0$$

$$\Rightarrow (x - 1) + 0 + (z - 0) = 0$$

$$\Rightarrow x - 1 + z = 0$$

$$\Rightarrow \underline{x + z - 1 = 0}.$$

$$H: a_1 x_1 + a_2 x_2 + \dots + a_n x_n + a_{n+1} = 0 \quad \text{Hyperplane.}$$

$$P_H^\perp P = P - \frac{a^T \cdot P + a_{n+1}}{|a|^2} \cdot a$$

Here,  $H: a_1 x + a_2 y + a_3 z + a_4 = 0$

$$1 \cdot x + 0 \cdot y + 1 \cdot z - 1 = 0$$

$$\Rightarrow (1, 0, 1) = a = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow -1 = a_{n+1}$$

$$\Rightarrow (1, 1, -1) = A.$$

$$H = P_H^\perp_{(BCD)} A = A - \frac{-1}{2} \cdot a$$

$$= A + \frac{1}{2} \cdot a$$

$$= \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} + \frac{1}{2} \cdot \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 + \frac{1}{2} \\ 1 \\ -1 + \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{3}{2} \\ 1 \\ -\frac{1}{2} \end{bmatrix} \Rightarrow H \left( \frac{3}{2}, 1, -\frac{1}{2} \right)$$

orthogonal proj.

⑥



$$\text{Ref}_{(BCD)}^\perp(A) = 2H - A$$

$$A' = 2 \cdot \begin{bmatrix} \frac{3}{2} \\ 2 \\ 1 \\ -\frac{1}{2} \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

$\Rightarrow A'(2, 1, 0)$  orthogonal reflection of A in (BCD).

$$P7) \text{ w.r. to } K, (\vec{AB} \times \vec{AC}) \times K + (\vec{AC} \times K) \times \vec{AB} = (*)$$

$$\vec{AB} = B - A = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\vec{AC} = C - A = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$$K = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\vec{AB} \times \vec{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 1 \\ -1 & 1 & 0 \end{vmatrix} = 1 \cdot (-1)^5 \cdot \begin{vmatrix} \hat{i} & \hat{j} \\ -1 & 1 \end{vmatrix} = -(\hat{i} + \hat{j}) = (-1, -1, 0)$$

$$(\vec{AB} \times \vec{AC}) \times K = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & -1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1 \cdot (-1)^6 \cdot \begin{vmatrix} \hat{i} & \hat{j} \\ -1 & -1 \end{vmatrix} = -\hat{i} + \hat{j} = (-1, 1, 0)$$

$$(\vec{AC} \times K) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = \cancel{(-1, 1, 0)} = 1 \cdot (-1)^6 \cdot \begin{vmatrix} \hat{i} & \hat{j} \\ -1 & 1 \end{vmatrix} = (\hat{i} + \hat{j}) = (1, 1, 0)$$

$$(\vec{AC} \times K) \times \vec{AB} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1 \cdot (-1)^6 \cdot \begin{vmatrix} \hat{i} & \hat{j} \\ 1 & 1 \end{vmatrix} = \hat{i} - \hat{j} = (1, -1, 0)$$

$$(*) = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$



18)  $K' \rightarrow$  left/right oriented?

$$K': (\vec{B}, \vec{BA}, \vec{AC}, \vec{CD}).$$

$$\vec{BA} = A - B = (0, 0, -1)$$

$$\vec{AC} = C - A = (-1, 1, 2)$$

$$\vec{CD} = D - C = (1, -2, -1)$$

$$\Rightarrow M = \begin{bmatrix} 0 & -1 & 1 \\ 0 & 1 & -2 \\ -1 & 2 & 1 \end{bmatrix}$$

$\downarrow \quad \downarrow \quad \downarrow$   
 $\vec{BA} \quad \vec{AC} \quad \vec{CD}$

$$\det M = \begin{vmatrix} 0 & -1 & 1 \\ 0 & 1 & -2 \\ -1 & 2 & 1 \end{vmatrix} = (-1) \cdot (-1)^4 \cdot \begin{vmatrix} -1 & 1 \\ 1 & -2 \end{vmatrix} = - (2 - 1) = -1 < 0$$

$\Rightarrow K'$  has opposite orientation to  $K$  (r-or)

$\Rightarrow K'$  is l-or.

19) w.r. to  $K'$ ; eq. of plane which contains  $\triangle ACD$ .

$$K': (\vec{B}, \vec{BA}, \vec{AC}, \vec{CD}).$$

$$K': (x', y', z')$$

$$\vec{BX} = x' \cdot \vec{BA} + y' \cdot \vec{AC} + z' \cdot \vec{CD}$$

$$\vec{BA} = A - B = (0, 1, -1)$$

$$\Rightarrow (0, 1, -1) = x' \cdot (0, 0, -1) + y' \cdot (-1, 1, 2) + z' \cdot (1, -2, -1)$$

$$\Rightarrow (0, 1, -1) = (-y' + z', y' - 2z', -x' + 2y' - z')$$

$$\Rightarrow \begin{cases} 0 = -y' + z' & \Rightarrow y' = z' \\ 1 = y' - 2z' & \Rightarrow 1 = -z' \Rightarrow z' = -1 = y' \\ -1 = -x' + 2y' - z' & \Rightarrow x' = z' + 1 = 0. \end{cases}$$

$$\Rightarrow (0, -1, -1) \Rightarrow [A]_{K'} = (0, -1, -1)$$

$$\vec{BC} = C - B = (-1, 2, 1) = -\vec{CD} = -(1, -2, -1) \cdot (-1, 1, 1).$$

$$\Rightarrow (-1, 2, 1) = x' \cdot (0, 0, -1) + y' \cdot (-1, 1, 2) + z' \cdot (1, -2, -1)$$

$$\Rightarrow (-1, 2, 1) = (-y' + z', y' - 2z', -x' + 2y' - z')$$

$$\Rightarrow \begin{cases} -1 = -y' + z' \\ 2 = y' - 2z' \\ 1 = -x' + 2y' - z' \end{cases}$$

$$\vec{BD} = (0, -1, 0).$$

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